Ch 9.2: Support Vector Classifier

Lecture 27 - CMSE 381

Michigan State University
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Dept of Computational Mathematics, Science & Engineering

Wed, Apr 1, 2026

Announcements

Last time:

- 9.1 Maximal Margin Classifier

This lecture:

- 9.2 Support Vector Classifier

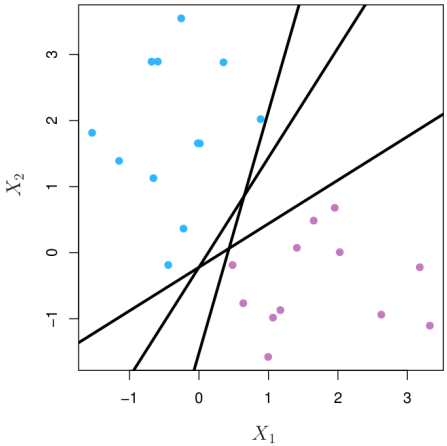
Announcements:

21	W	3/18	Polynomial & Step Functions	7.1-7.2		
22	F	3/20	Step Functions; Basis functions; Start Splines	7.2-7.4		
23	M	3/23	Regression Splines	7.4		
24	W	3/25	Decision Trees	8.1		Q7
25	F	3/27	Random Forests	8.2.1, 8.2.2	HW #5 Due Sun 3/29	
26	M	3/30	Maximal Margin Classifier	9.1		
27	W	4/1	SVC	9.2		Q8
28	F	4/3	SVM	9.3, 9.4		
29	M	4/6	Single Layer NN	10.1		
30	W	4/8	Multi Layer NN	10.2		Q9
31	F	4/10	CNN	10.3	HW #6 Due Sun 4/12	
32	M	4/13	Unsupervised learning / clustering	12.1, 12.4		Q10
33	W	4/15	Virtual: Project Office Hours			
	F	4/17	***Review***			
	M	4/20	**Midterm #3**			
	W	4/22				
	F	4/24			**Project Due**	

Section 1

Last time

Separating Hyperplane



Require that for every data point:

$$\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} > 0 \text{ if } y_i = 1$$
$$\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} < 0 \text{ if } y_i = -1$$

Equivalently

Require that for every data point

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) > 0$$

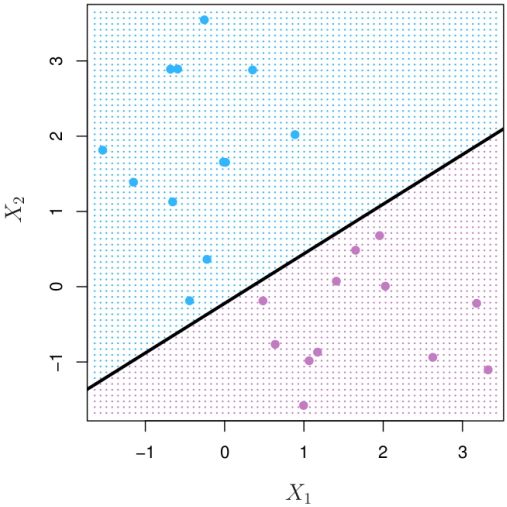
Separating hyperplane becomes a classifier

If you have a separating hyperplane:

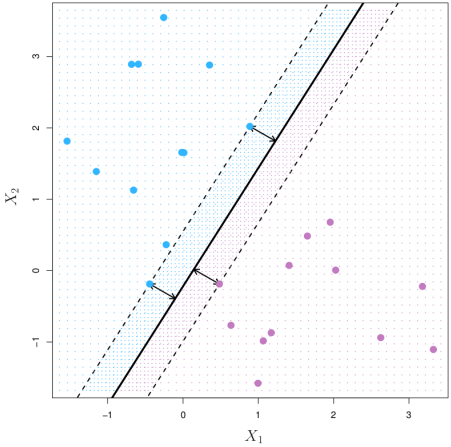
- Check

$$f(x^*) = \beta_0 + \beta_1 x_1^* + \beta_2 x_2^* + \cdots + \beta_p x_p^*$$

- If positive, assign $\hat{y} = 1$
- If negative, assign $\hat{y} = -1$



Maximal margin classifier



- For a hyperplane, the *margin* is the smallest distance from any data point to the hyperplane.
- Observations that are closest are called *support vectors*.
- The *maximal margin hyperplane* is the hyperplane with the largest margin
- The classifier built from this hyperplane is the *maximal margin classifier*.

Test your understanding: PollEv

Mathematical Formulation

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,M}{\text{maximize}} \quad M$$

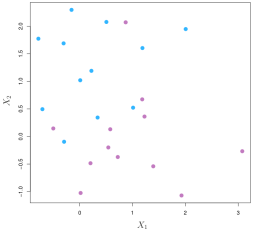
$$\text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 = 1,$$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M \quad \forall\, i = 1, \ldots, n$$

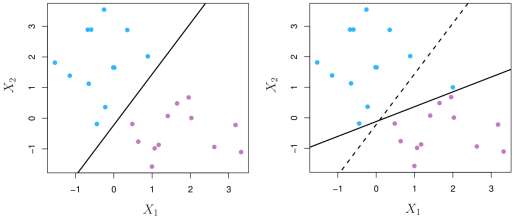
- Unit normal requirement, we can always write a given hyperplane this way.
- Last eq forces points to be on the correct side of the hyperplane in order for $M > 0$
- The product is the distance from the point to the hyperplane
- Making M as big as possible is the maximal margin hyperplane

Problems

Might be no separating hyperplane



Sensitivity to new points



What will you learn today?

- What is the difference between Support Vector Classifier and Maximal Margin Classifier?
  - What short-comings of Maximal Margin Classifier does Support Vector Classifier overcome?
  - How?
  - You should be able to describe the difference verbally, graphically, and mathematically.
- How to interpret each element of the mathematical formulation of Support Vector Machine graphically?
  - Given a graph of Support Vector Classifier and some data points, you should be able to derive the values of different parameters in the mathematical formulation (e.g. M, ϵ_i).
- What parameters controls the flexibility (hence bias-variance trade-off) of Support Vector Classifier?
- What are the support vectors of Support Vector Machine?

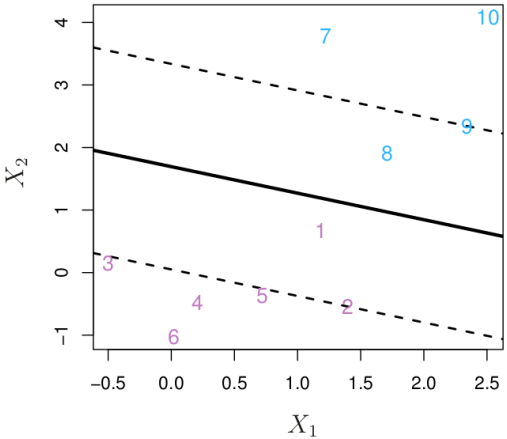
Section 2

Support Vector Classifier

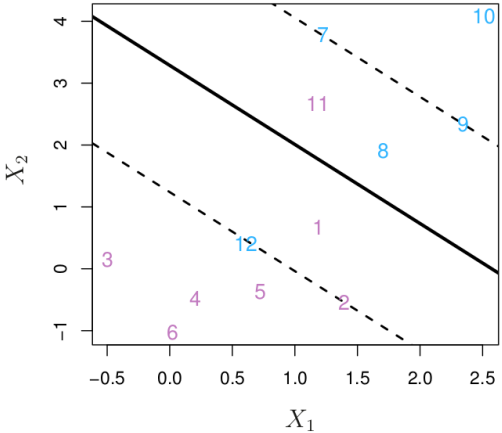
Basic idea

- Be ok with having a classifier that isn't quite perfect
- Aim for greater robustness to individual observations
- Better classification of most of the training observations
- Result is a support vector classifier
- Soft margin classifier

Soft margin



Some points on wrong side of margin



Some points on wrong side of hyperplane (Misclassified)

Mathematical Formulation of SVC

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,\epsilon_1,\ldots,\epsilon_n,\,M}{\text{maximize}} \quad M$$

$$\text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 = 1,$$

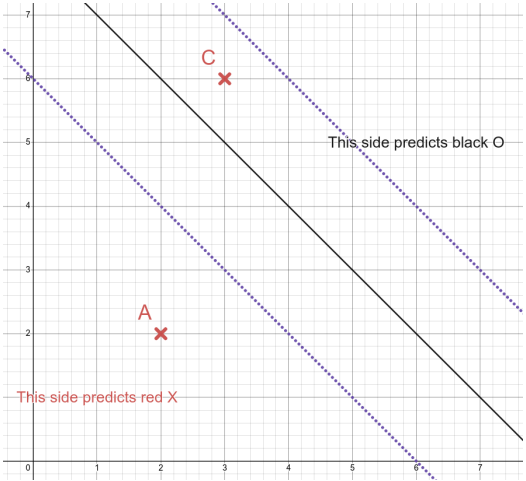
$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M(1 - \epsilon_i),$$

$$\epsilon_i \geq 0, \quad \sum_{i=1}^{n} \epsilon_i \leq C,$$

- C is nonnegative tuning parameter
- M is the width of the margin
- $\varepsilon_1, \cdots, \varepsilon_n$ are slack variables allowing observations to go to the other side

Find positive ε's that will satisfy this

Fix $M = \sqrt{2}$ $\qquad y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M(1 - \varepsilon_i)$

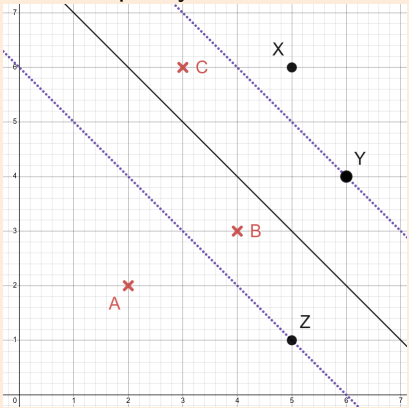


- Point out M is distance from center hyperplane to noted margin.
- A is on the correct side of the hyperplane.
  - Left side is already bigger than M, so set $\varepsilon_i = 0$.
- C is on the wrong side of the hyperplane for its label.
  - Left side of the equation is negative. What ε satisfies this?
  - $-\frac{\sqrt{2}}{2} = \sqrt{2}(1 - \varepsilon)$
  - $\varepsilon = 3/2$

What is ε?

Fix $M = \sqrt{2}$ $\qquad y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M(1 - \varepsilon_i)$

Fill in the table so that the inequality is satisfied.

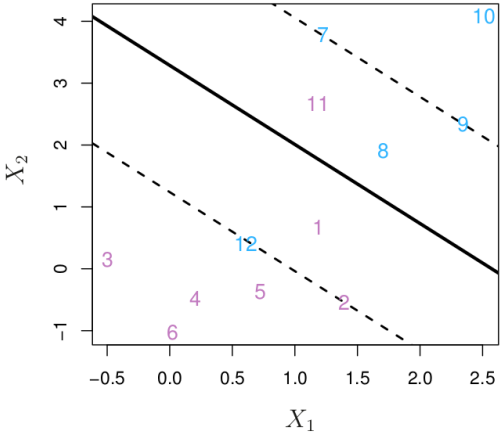


Point	*Left Side*	ε_i	$M(1-\varepsilon_i)$
A	$2\sqrt{2}$	*0*	$\sqrt{2}$
B	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	$\sqrt{2}(1-1/2) = \frac{\sqrt{2}}{2}$
C	$-\frac{\sqrt{2}}{2}$	1.5	$-\frac{\sqrt{2}}{2}$
X	$\frac{3\sqrt{2}}{2}$	*0*	$\sqrt{2}$
Y	$\sqrt{2}$	*0*	$\sqrt{2}$
Z	$-\sqrt{2}$	*2*	$\sqrt{2}(1-2) = -\sqrt{2}$

To use for later slide: Total ε is 4

What is ε?

- If $\varepsilon_i = 0$, then on correct side of margin
- If $\varepsilon_i > 0$ then on the wrong side of margin (Violated margin)
- If $\varepsilon_i > 1$ then on the wrong side of hyperplane



What is C?

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,\epsilon_1,\ldots,\epsilon_n,\,M}{\text{maximize}} \quad M$$

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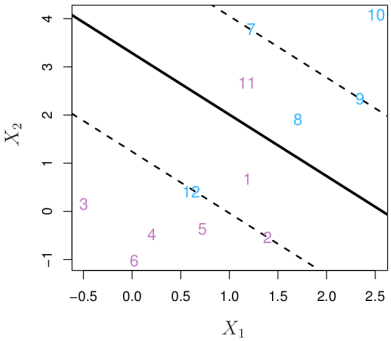
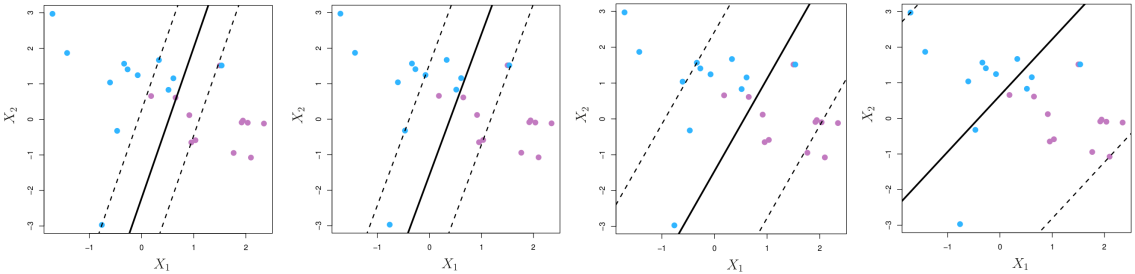
$$\epsilon_i \geq 0, \quad \sum_{i=1}^{n} \epsilon_i \leq C,$$

- Bounds sum of ε_i, so controls number & severity of violating margin (budget)
- $C = 0$ means no violations allowed
- $C > 0$ means at most C observations can be on wrong side of hyperplane
- In previous example, our total of ε was 4, so would be a valid hyperplane for C at most 4.



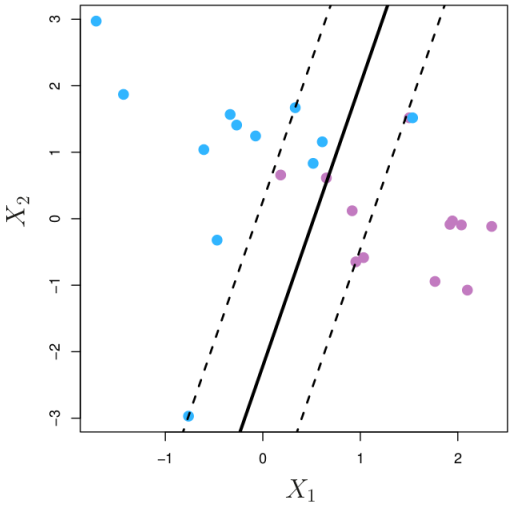
Examples messing with C



Increasing $C \rightarrow$

- For increasing C, we have a wider margin, so more points allowed violate margin/hyperplane
- C controls bias-variance trade-off
- small C: highly fit to the data, low bias, high variance
- large C: fitting less hard, high bias, low variance

What affects the hyperplane?



- Only observations on the margin or violating the margin affect the hyperplane
- These observations are called support vectors
- Changing other points positions doesn't affect hyperplane found
-

Coding

TL;DR

Maximal Margin Classifier

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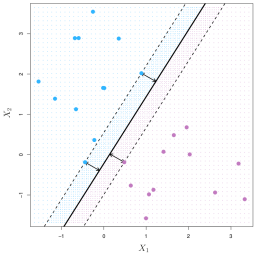

Support Vector Classifier

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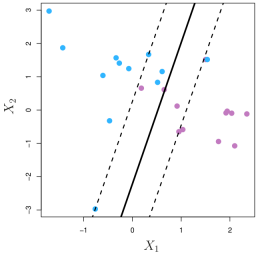

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